HTE LITE

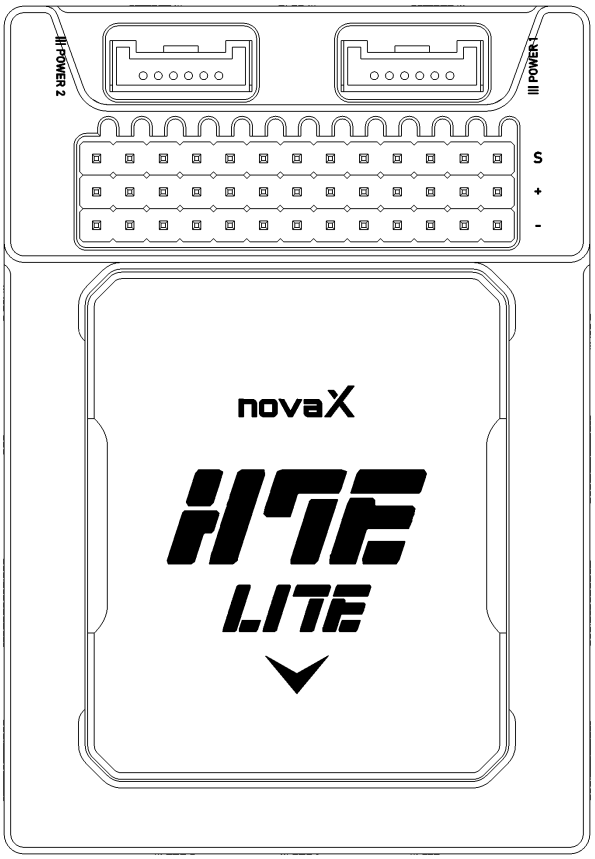
User Guide



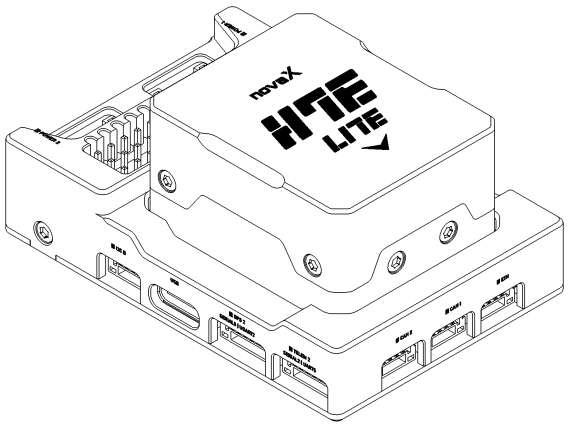
novaX

1. Parts List

· Main & Board



H7E Lite Autopilot



1. Parts List

· Connector Type

Port	Connector
POWER 1 · POWER 2	Molex Micro-Lock Plus 6P (1.25 mm)
TELEM 1 · TELEM 2 · GPS 1 · GPS 2	JST-GH 6P
SERIAL4 · SERIAL6 · I2C A · I2C B · CAN 1 · CAN 2 · ETHERNET	JST-GH 4P
RC IN	JST-GH 5P
DEBUG	JST-SH 6P
USB	USB Type-C
PWM OUT M1–M12 · SB (SBUS OUT)	2.54 mm 3 × 13 pin header

2. Specifications

· Hardware Specifications (1)

MCU	STM32H753, ARM Cortex-M7, 480 MHz
RAM / Flash	1 MB / 2 MB
IMU	2× ICM-42688-P + BMI088
IMU Heater	Built-in heater on the IMU board, held at 45 °C
Magnetometer	RM3100
Barometer	2× ICP-20100
Operating Voltage	4.75 – 5.7 V (Rated 5 V)
USB Input	4.75 – 5.25 V
Power Input	POWER 1 + POWER 2 (redundant, I2C power modules)
PWM Output	12 FMU channels (DShot except M7–M8) + SBUS out
RC Input	S.Bus, PPM, DSM / Spektrum
RSSI Input	Analog
UART	6 serial ports (TELEM 1 · 2, GPS 1 · 2, SERIAL4, SERIAL6) + debug console
GPS	GPS 1 + GPS 2 (6-pin, UART + I2C, left / right)

2. Specifications

· Hardware Specifications (2)

I²C	2 dedicated ports (I2C A, I2C B) + GPS 1 · 2 + power-module bus on POWER 1 · 2
CAN	2 Port
Ethernet	100 Mbps x 1 Port
Size	44.5 × 64.5 × 28.6 mm
Mount Hole	16 × 30 mm / M3 (4 blind holes in the base, 3 mm thread)
Supported F/W	novaX ArduPilot

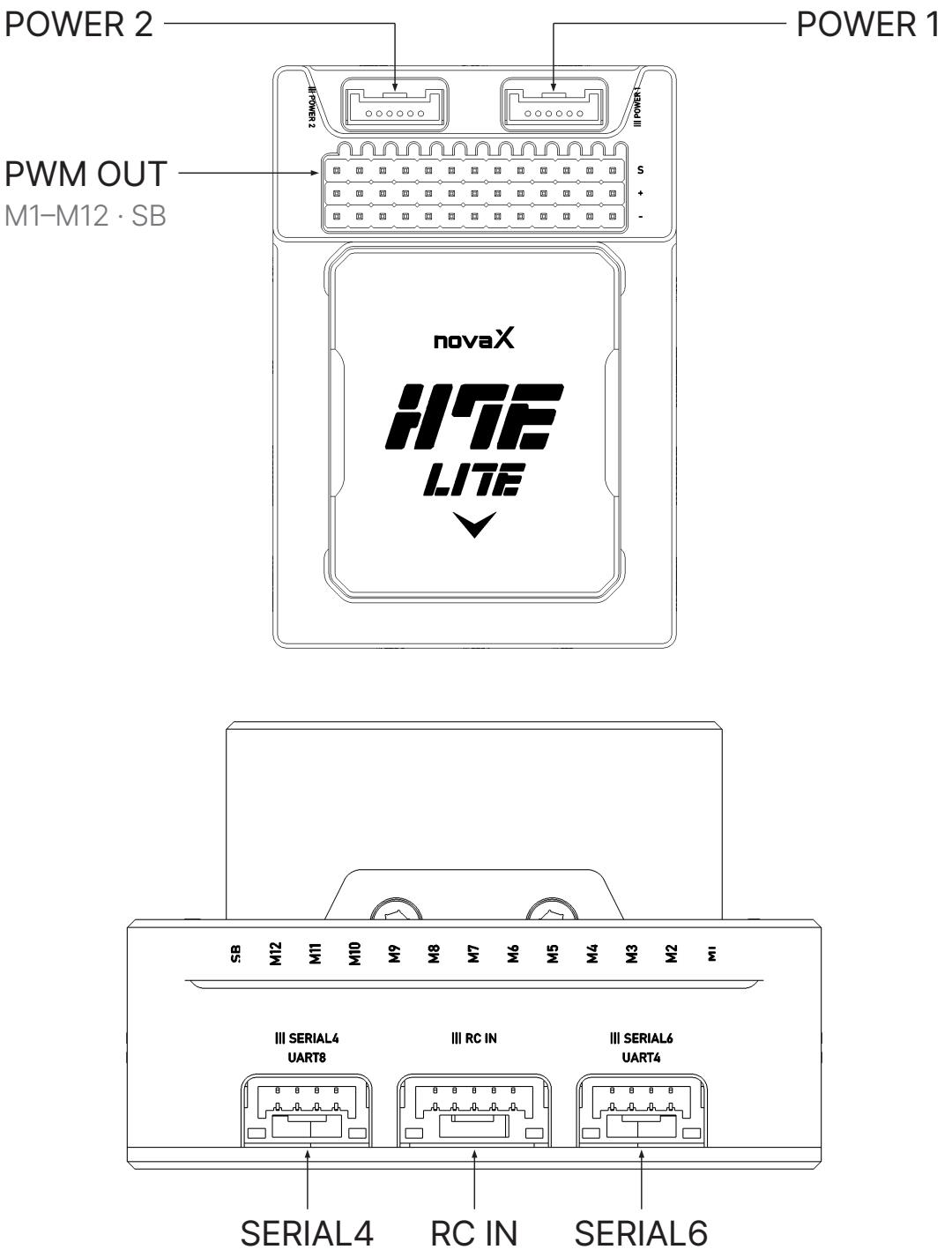
2. Specifications

· Interfaces

POWER	2	ETHERNET	1
TELEM	2	DEBUG	1
GPS	2	USB	1
SERIAL	2	PWM Outputs	12
I2C	2	SBUS OUT	1
CAN	2	TF Card Slot	1
RC IN	1		
Power Module	POWER 1 + POWER 2 (redundant, I2C power modules)		

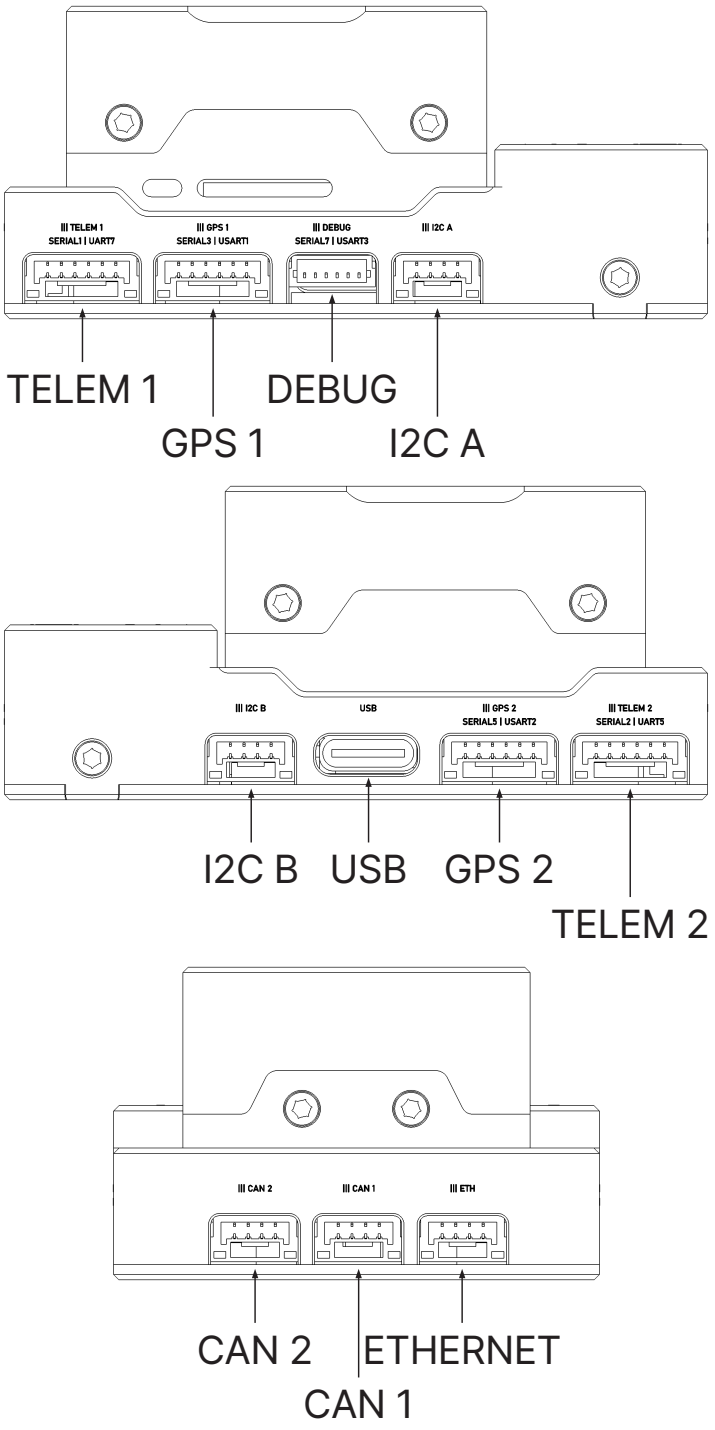
3. Pinouts

· Pinmap (1)



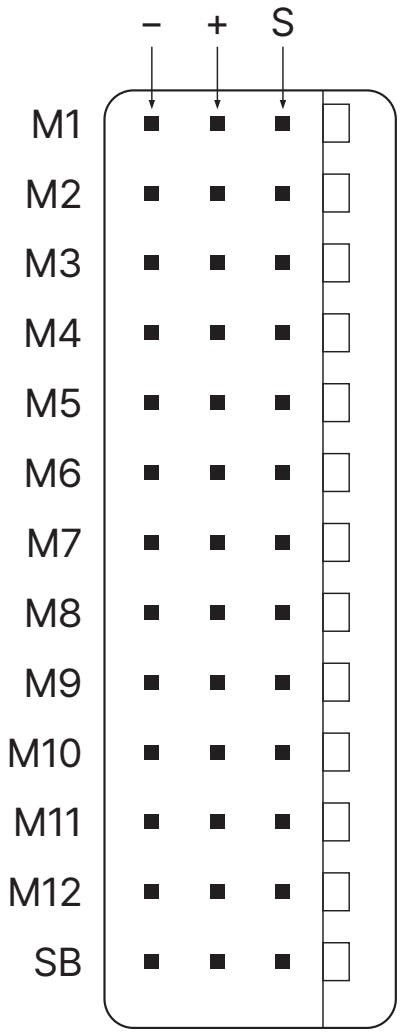
3. Pinouts

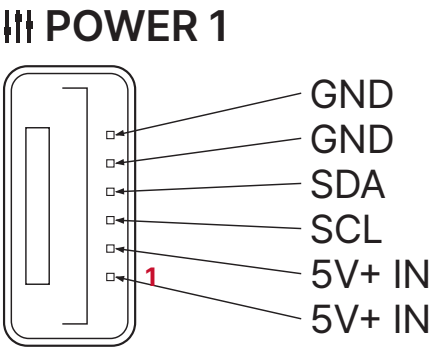
· Pinmap (2)

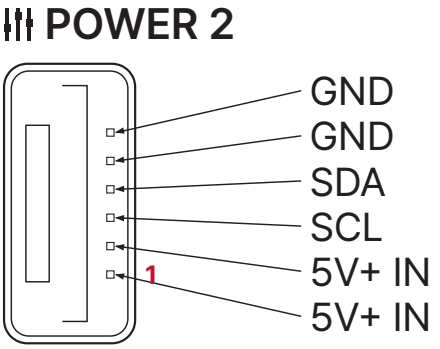


3. Pinouts

· Top





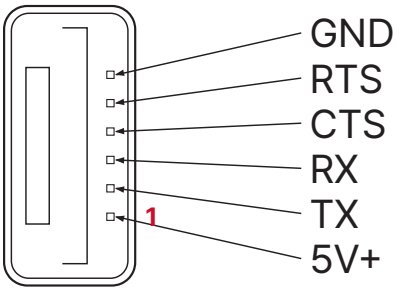


SB = SBUS OUT · + = servo rail (external)

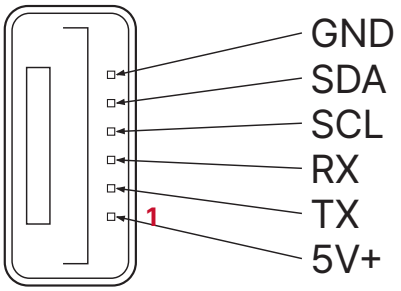
3. Pinouts

· Left

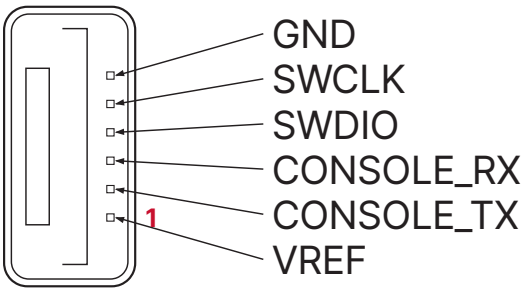
TELEM 1



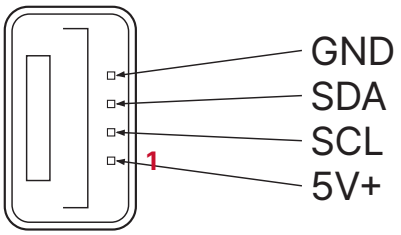
GPS 1



DEBUG



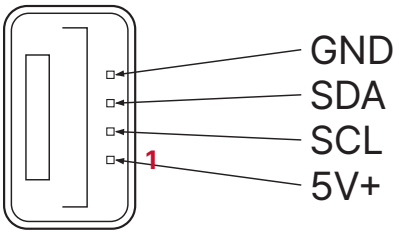
I2C A



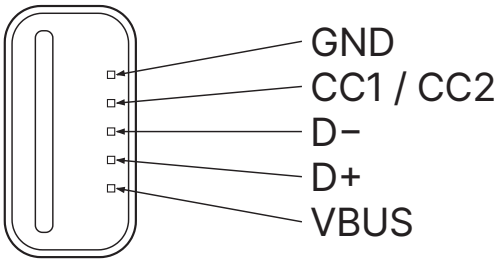
3. Pinouts

· Right

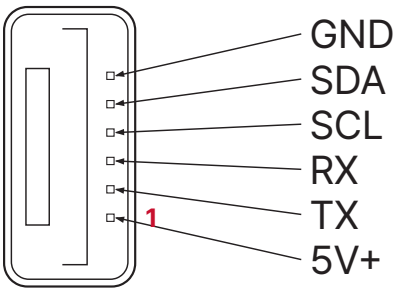
I2C B



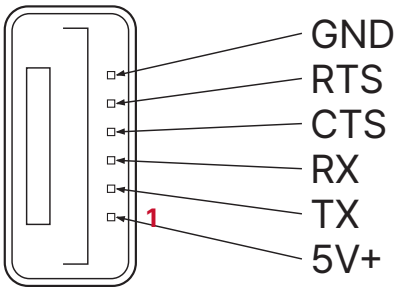
USB



GPS 2



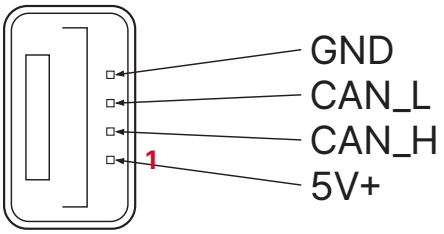
TELEM 2



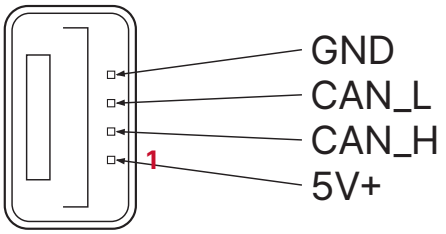
3. Pinouts

· Front · Rear

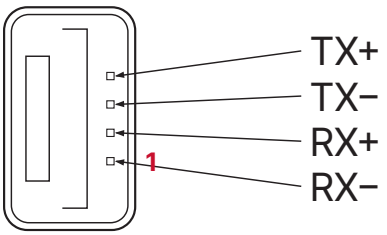
CAN 2



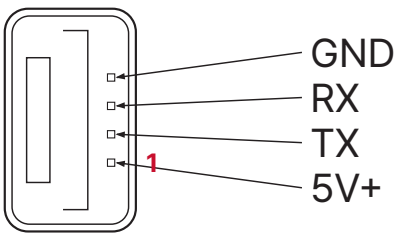
CAN 1



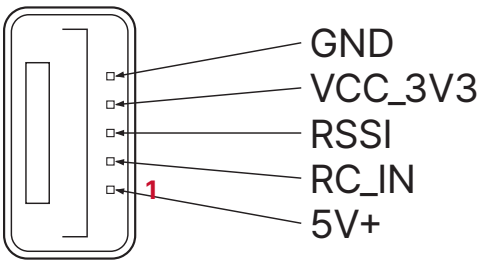
ETHERNET



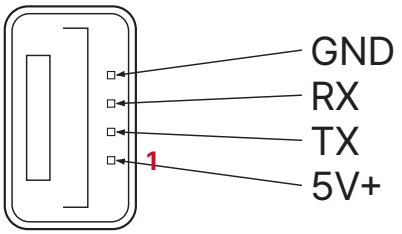
SERIAL4



RC IN



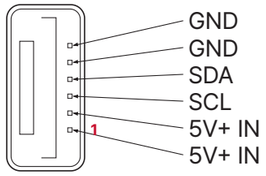
SERIAL6



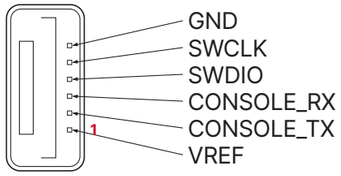
novaX

HTE LITE

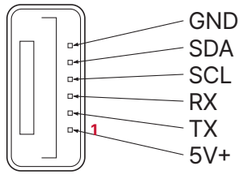
POWER 1



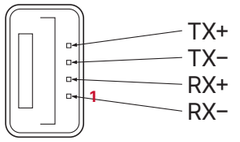
DEBUG



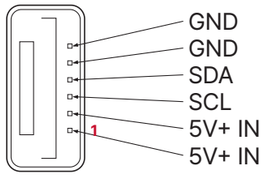
GPS 2



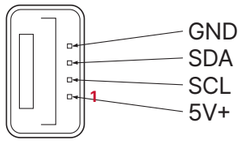
ETHERNET



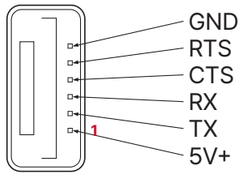
POWER 2



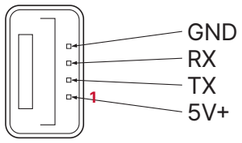
I2C A



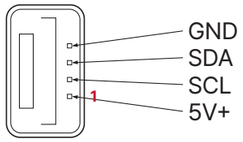
TELEM 2



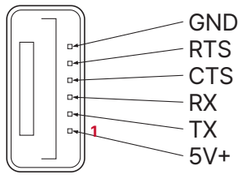
SERIAL4

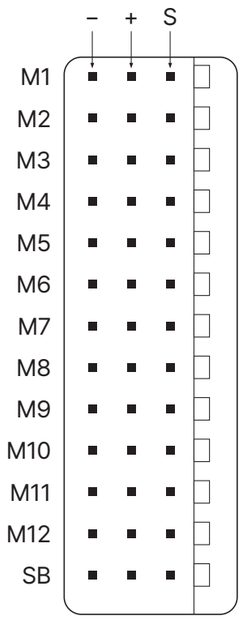


I2C B

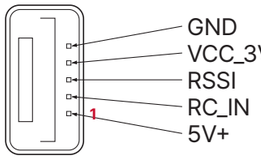


TELEM 1

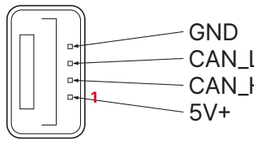




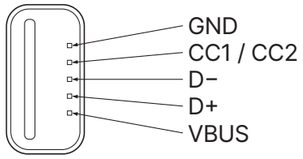
RC IN



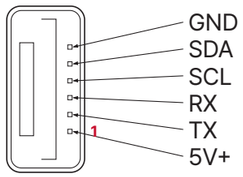
CAN 2



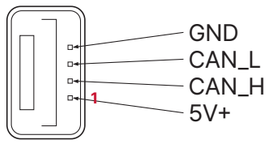
USB



GPS 1



CAN 1



SERIAL6

